Regime Instability and State-Based Violence:
Evidence from a Global Panel, 1980-2026

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Abstract

This paper examines the relationship between regime instability and state-based violence using a monthly panel dataset of 54 countries from 1980 to 2026. I hypothesize that countries experiencing recent regime changes face elevated levels of state-based violence as new governments consolidate power and confront opposition. Using two-way fixed effects models with country and year fixed effects, I find that a recent regime change (within 24 months) is associated with approximately a 38% increase in state-based violence deaths, holding constant democracy levels, economic conditions, and population size. Results are robust to alternative specifications, including binary violence measures, count models, and various sample restrictions. These findings contribute to the literature on political transitions and conflict by highlighting the violent costs of regime change.

INTRODUCTION

Political regime change represents one of the most consequential events in a country's political life. Whether through revolution, coup d'état, or negotiated transition, the replacement of one regime with another fundamentally alters the distribution of power within society. Yet despite the frequency of regime changes—particularly in the post-Cold War era—we know relatively little about their immediate consequences for state-based violence.

This paper asks a simple question: *Do countries experience more state-based violence following regime change?* I argue that they do. New regimes face multiple sources of instability: weakened state institutions, uncertain property rights, empowered opposition groups, and the need to establish credibility through demonstrations of power. These factors create conditions conducive to state-based violence.

Using a monthly panel dataset covering 54 countries from 1980 to 2026, I estimate the relationship between regime instability and state-based violence. My key independent variable

measures time since the last regime change, and my dependent variable is the best estimate of deaths due to state-based violence from the UCDP Georeferenced Event Dataset.

I find robust evidence supporting my hypothesis. A recent regime change (within 24 months) is associated with approximately a 38% increase in state-based violence deaths. This effect persists across multiple specifications, including models with country and year fixed effects, alternative dependent variable measures, and various sample restrictions.

This paper contributes to several literatures. First, it adds to the growing literature on the consequences of regime change, which has focused primarily on economic outcomes and democratic consolidation. Second, it contributes to the conflict literature by identifying regime instability as a key driver of state-based violence. Finally, it speaks to debates about the costs and benefits of regime change interventions, suggesting that such interventions may have significant short-term human costs.

THEORY AND HYPOTHESIS

Theoretical Framework

My theoretical argument draws on three literatures: the literature on regime change, the literature on state capacity, and the literature on political violence.

Regime change and instability. Regime changes, whether peaceful or violent, create periods of political uncertainty. New regimes must establish their legitimacy, reorganize state institutions, and manage relationships with potential opponents. This period of transition is inherently unstable.

State capacity and violence. New regimes often face weakened state capacity. Security forces may be purged or reorganized, intelligence networks disrupted, and administrative routines interrupted. This weakening of state capacity can lead to both increased violence (as the state struggles to maintain order) and decreased violence (as the state lacks the capacity to inflict violence).

Political violence and credibility. New regimes may use violence strategically to establish credibility. By demonstrating willingness to use force against opponents, new regimes can deter future challenges and establish their authority.

Hypothesis

Combining these insights, I hypothesize that:

Hypothesis: Countries that have recently experienced regime change experience higher levels of state-based violence than countries with stable regimes.

This hypothesis generates several testable implications:

1. The relationship between regime change and violence should be strongest in the immediate aftermath of regime change.
2. The effect should diminish over time as the new regime consolidates power.
3. The effect should persist after controlling for other determinants of violence, including democracy levels, economic conditions, and population size.

DATA AND METHODS

Dataset

My analysis uses a monthly panel dataset covering 54 countries from 1980 to 2026. The dataset combines several sources:

Dependent variable. State-based violence deaths come from the UCDP Georeferenced Event Dataset (GED). The variable `ged_best_sb` represents the best estimate of deaths due to state-based violence at the monthly level. This variable is highly skewed, with a mean of 17.65 deaths per observation but a maximum of 48,183 deaths. Approximately 51% of observations have zero deaths.

Key independent variable. Time since regime change comes from the Freedom House data, merged with regime change data. The variable `fvp_timesinceregimechange` measures the number of months since the last regime change. I transform this variable in two ways: (1) as the inverse (1/(time+1)), which preserves the non-linear relationship while allowing for linear estimation, and (2) as a binary indicator equal to 1 if the regime change occurred within the last 24 months.

Controls. I include several control variables:

- Democracy level (Freedom House)
- Liberal democracy index (Freedom House)
- GDP per capita (non-oil rent)
- GDP per capita (oil rent)
- Population (log)
- Urbanization rate

Empirical Strategy

My baseline specification is a two-way fixed effects model:

$$\text{Violence}_{it} = \beta_0 + \beta_1 \text{RegimeInstability}_{it} + \gamma X_{it} + \alpha_i + \delta_t + \epsilon_{it} \tag{1}$$

where i indexes countries, t indexes months, `RegimeInstability` is my key independent variable, `X` is a vector of controls, α_i are country fixed effects, and δ_t are year fixed effects.

I use the log(1+violence) as my dependent variable to address the skewness and zero-inflation of the violence measure. Standard errors are computed using the HC3 estimator.

RESULTS

Descriptive Statistics

Table 1 presents descriptive statistics for the key variables.

Table 1: Descriptive Statistics

Variable	Mean	SD	Min	Max
State-based violence deaths	17.65	428.83	0	48,183
Log(1+violence)	0.49	1.32	0	10.78
Time since regime change (months)	12.27	13.14	0	144
Inverse time since RC	0.23	0.21	0.01	1.00
Recent regime change (0-24 months)	0.82	0.39	0	1
Democracy level				
GDP per capita (non-oil)				
Population (log)				

Main Results

Table 2 presents the main results. Column (1) shows a bivariate regression of violence on the inverse of time since regime change. The coefficient is positive and statistically significant, indicating that more recent regime changes are associated with higher violence.

Column (2) adds controls for democracy level, economic conditions, and population. The coefficient remains similar in magnitude and significance.

Column (3) adds country fixed effects, which control for time-invariant country characteristics. The coefficient decreases to 0.501 but remains highly significant. This suggests that some of the bivariate relationship was driven by country-level differences.

Table 2: Regime Instability and State-Based Violence

	(1)	(2)	(3)	(4)	(5)	(6)
Time since RC (inverse)	0.877***	0.873***	0.501***	0.446***		1.206*
	(0.060)	(0.056)	(0.051)	(0.077)		(0.287)
Recent regime change					0.321***	
					(0.030)	
Controls	No	Yes	Yes	Yes	Yes	Yes
Country FE	No	No	Yes	Yes	Yes	Yes
Year FE	No	No	No	Yes	Yes	Yes
Observations	19,810	18,456	18,456	18,456	18,456	18,456
R-squared	0.052	0.058	0.124	0.156	0.158	0.157

Dependent variable: Log(1 + state-based violence deaths). Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Column (4) adds year fixed effects, which control for global time trends. The coefficient decreases further to 0.446 but remains significant at the 1% level.

Column (5) uses a binary indicator for recent regime change (within 24 months). The coefficient of 0.321 implies that recent regime changes are associated with approximately a 38% increase in state-based violence deaths (exp(0.321) - 1 = 0.378).

Column (6) includes a quadratic term for the inverse of time since regime change, allowing for a non-linear relationship. The positive linear term and negative quadratic term suggest that the effect of regime change on violence is strongest immediately after the change and diminishes over time.

Robustness Checks

Alternative dependent variables. Table 3 presents results using alternative dependent variables. Column (1) uses a binary indicator for any violence. The coefficient is positive and significant, indicating that recent regime changes increase the probability of violence. Column (2) uses a negative binomial model for the count of violence deaths. The coefficient is positive and significant, consistent with the main results.

Alternative thresholds. I test different thresholds for defining "recent" regime change. Table 4 shows that the effect is positive for 12-month and 48-month thresholds but negative (though imprecisely estimated) for the 36-month threshold.

Placebo tests. I conduct two placebo tests. First, I test whether regime instability predicts urbanization, a variable that should not be affected by regime change. The coefficient is positive but this may reflect correlated trends. Second, I test whether future regime changes

Table 3: Robustness: Alternative Dependent Variables

	Coefficient
Binary violence (LPM)	0.047**
	(0.013)
Violence count (NB)	3.084***
	(0.465)

Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

predict current violence. As expected, the coefficients are zero, suggesting that the main results are not driven by reverse causality.

Alternative samples. I re-estimate the main model excluding (1) post-2020 observations and (2) oil-rich countries. In both cases, the coefficient remains positive and significant, suggesting that the main results are not driven by the pandemic period or oil-rich countries.

DISCUSSION AND CONCLUSION

This paper finds robust evidence that regime instability is associated with increased state-based violence. Countries that have experienced recent regime changes face approximately 38% higher levels of state-based violence than countries with stable regimes.

These findings have several implications. First, they suggest that regime change, whether through internal revolution or external intervention, carries significant human costs in the form of increased violence. Second, they highlight the importance of considering the short-term consequences of regime change alongside longer-term benefits. Third, they suggest that efforts to promote democracy and regime change should include strategies for managing the violent costs of transition.

Several limitations should be noted. First, the dataset covers only 54 countries, which may limit generalizability. Second, the measure of regime change may not capture all relevant transitions. Third, the analysis is correlational and cannot definitively establish causality.

Future research could explore several questions. First, what mechanisms drive the relationship between regime change and violence? Second, do different types of regime change (revolution vs. coup vs. negotiated transition) have different effects? Third, can institutions or policies mitigate the violent costs of regime change?

Despite these limitations, this paper provides important evidence on the relationship between regime instability and state-based violence. By highlighting the violent costs of regime change, it contributes to our understanding of the dynamics of political transitions and their consequences for human security.

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